IQREDO

Fabric IQ in two hours

From readings to decisions: an ontology, a graph and three agents on a bike-share control room

Brian Bønk · Fabric IQ redo · Nov 10, 2026

Two hours, seven labs

Time	Block	You build
0:00	Kick-off	–
0:10	01 Prepare the data	Lakehouse + Eventhouse
0:25	02 Build the ontology by hand	6 entity types, 6 relationships
0:50	03 Give it business meaning	Descriptions, synonyms, rules
1:02	04 Explore the graph	Graph + path query
1:15	05 Ontology agent	Hero question; Plan → Act
1:30	06 Data agent	BikeOpsDataAgent
1:45	07 Operations agent	An incident, a Teams message
1:55	Wrap-up	–

Everything by hand in the Fabric portal · all data as CSV · github.com/brianbonk/IQRedo

THE GAP

Correct data, no decision

What the data says (K5)

BikepointID	EmptyMinutes
BP-001	38
BP-061	31
BP-003	24
BP-064	19
BP-071	7

What the control room needs

“York Way in Kings Cross, a Tier 1 area, has been empty for 38 minutes. That’s North Loop’s patch: their van leaves Kings Cross Depot. Take V01 with 14 bikes; V02 is off the road.”

Who do you call?

The answer isn’t in any table. It’s in how the business is organised.

WHAT AN ONTOLOGY IS

A shared model of the business

Things

Entity types: Station, Crew, Van

Identity

Keys: what makes a row one real station

Connections

Relationships: a station is located in a neighbourhood

Meaning

Descriptions, synonyms and business rules

Bindings

Where the data for each thing lives

Not a copy

The data stays in OneLake

HOW IT FITS TOGETHER

Data below, meaning in the middle, agents on top

Ontology agent
for the people who build the model

Data agent
answers for everybody

Operations agent
watches and recommends

BikeOpsOntology

entity types · keys · relationships · descriptions · rules · bindings

+ optional graph (snapshot)

Lakehouse · master data
stations, neighbourhoods, crews, depots, vans

Eventhouse · readings
one row per station per minute

WHERE FABRIC IQ SITS

RTI reacts to signals. IQ adds what they mean.

	Real-Time Intelligence	Fabric IQ
Speaks	Tables and columns	Business concepts
Knows	BP-003 has 0 bikes	Argyle Street is North Loop’s patch
Rule	No_Bikes < 1 for 5 min	Needs rebalancing (Tier 1, > 15 min)
Acts by	Activator alert	Agents that explain and recommend
Built in	Eventstream, Eventhouse, Activator	Ontology, graph, agents

Not either–or: IQ sits on top of the data RTI brings in.

TODAY'S SCENARIO

A London bike-share control room

72
docking stations

12
neighbourhoods, Tier 1 or 2

6
crews, each with a depot

10
vans

Every station reports bikes and empty docks every minute: 12,960 readings over three hours, rebased so they end **now**.

BY 1:30 YOU'LL ASK ONE QUESTION

"Which Tier 1 neighbourhoods have stations that have been empty for more than 15 minutes? How long, how many bikes, which crew, which depot, which van?"

live readings | master data | 4 hops of relationships | business rules

LAB 01 · PREPARE THE DATA · 15 MIN

Put each kind of data where it belongs

Things → Lakehouse

Stations, neighbourhoods, crews, depots, vans. Change rarely. Managed Delta tables, plain column names.

`BikeOpsLakehouse`

Readings → Eventhouse

One row per station per minute. Time-series, append-only, queried with KQL.

`BikeOpsEventhouse`

Keep van_maintenance.csv on your laptop: Lab 05 needs it.

LAB 02 · BUILD THE ONTOLOGY · 25 MIN

Entity types, bindings and keys

Entity type

A kind of thing the business talks about. Station, Crew, Van.

Binding

Columns of a Lakehouse or Eventhouse table become its properties. Read at query time.

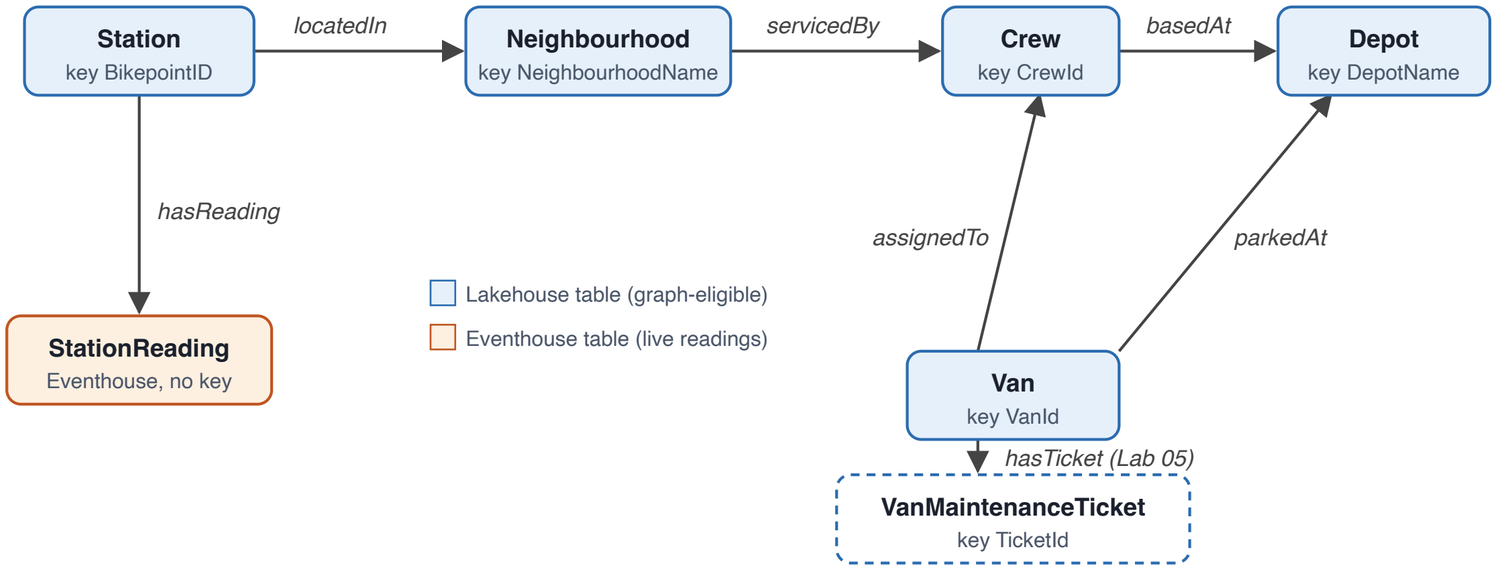
Key

What makes a row one real station. Needed for the graph.

StationReading binds to the Eventhouse and gets **no key**: one station has thousands of readings.

LAB 02 · RELATIONSHIPS

Every relationship is a join you never write again



Origin property matches target property · names must be unique · direction reads as a sentence: Van assignedTo Crew

LAB 03 · BUSINESS MEANING · 12 MIN

Agents only know what you write down

The schema says

```
No_Bikes: long
Priority: string
```

The business says

“Empty means nobody can rent a bike there. Tier 1 areas need someone on it within 15 minutes.”

Descriptions what it is

Synonyms dock, bike point

Rules what words mean

LAB 03 · BUSINESS RULES

Rules are definitions, not triggers

Rule	Definition (shortened)
Empty station	Latest reading has No_Bikes = 0
Needs rebalancing	Empty for more than 15 min, or Low in a Tier 1 neighbourhood
Responsible crew	The crew that services the station’s neighbourhood
Bikes needed	round(Capacity × 0.5) − No_Bikes
Dispatchable van	VanStatus is Available

Written in natural language, linked to entity types. Agents read them to ground answers; nothing fires.

LAB 04 · EXPLORE THE GRAPH · 13 MIN

Use the graph when the path is the answer

Optional

Opt in with Manage graph. Most questions go straight to the sources.

Lakehouse only

Entities need a key and a Lakehouse or mirrored source. StationReading stays out.

A snapshot

It doesn’t refresh by itself. Refresh now, or set a schedule.

Argyle Street → Kings Cross → North Loop → Kings Cross Depot

3 hops, no joins

LAB 05 · ONTOLOGY AGENT · 15 MIN

Let the agent type. You review.

Plan mode (default)

- Explain the ontology
- Query: SQL, KQL or GQL, chosen from the binding
- Propose a change and show a draft preview

Changes nothing.

Act mode

- Applies the plan you reviewed
- Says what it’s about to do first
- Needs Contributor on the workspace

Switch back to Plan right after.

LAB 05 · SAME QUESTION, BETTER ANSWER

Fix the model, not the prompt

Crew	Before maintenance tickets	After (agent added them)
North Loop Kings Cross, 32 bikes	V01 or V02	Only V01 (V02: open High ticket) → two trips
Night Owls Waterloo, 34 bikes	V09	No dispatchable van (V09: High ticket, V10 in service) → borrow from another crew

One new entity type, one relationship, one edited rule. Nobody changed the question.

LAB 06 · DATA AGENT · 15 MIN

Two agents, two jobs

	Ontology agent	Data agent
For	People who build the model	People who want answers
Lives	Inside the ontology item	Its own item, shareable
Does	Explain, query, propose, apply	Answer questions, read-only
Reaches	Fabric only	Copilot, Teams, Foundry
Ontology	Edits it	Reads it as context

LAB 06 · WHAT GOES WHERE

Meaning in the ontology, manners in the agent

Ontology

- What “empty” and “Tier 1” mean
- How a station finds its crew
- Which van is dispatchable

Shared by every agent

Agent instructions

- Names over IDs
- “Now” = the latest reading
- Table first, then three sentences

Specific to this agent

LAB 07 · OPERATIONS AGENT · 10 MIN

Same signal, a decision attached

RTI: Activator rule

```
No_Bikes < 1 for 5 minutes → email
```

“BP-043 has no bikes.” Someone still works out where, whose patch, which van.

IQ: operations agent

Watches through the ontology, writes a playbook from your goal.

“Bayswater Road, Hyde Park, empty 12 min. West Wheels from Paddington Depot, V06, 18 bikes.” **Yes / No?**

WRAP-UP

What you built in two hours

1. **Data where it belongs**: things in a Lakehouse, readings in an Eventhouse
2. **An ontology**: business names, keys, relationships instead of joins
3. **Business meaning**: descriptions, synonyms, rules, written once
4. **A graph**: multi-hop questions by following relationships
5. **Three agents**: one builds with you, one answers, one watches

When an answer is wrong, fix the model, not the prompt.

Before you leave

Stop BikeOpsOperationsAgent · delete the workspace when you’re done · labs and data: github.com/brianbonk/IQRedo